**5. Instrumentation**
└ **5.2 Alerting and metrics**
└ **5.3.2 Analyze app usage/performance metrics**

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1. What Azure tools are used to analyze application usage and performance metrics?
2. How do you enable application performance monitoring in Azure?
3. What types of application usage metrics can be collected in Azure?
4. How do you visualize and interpret performance metrics in Application Insights?
5. How do you configure custom metrics and events in Application Insights?
6. What is the difference between availability, performance, and usage metrics?
7. How do you set up alerts based on application performance thresholds?
8. How can you use Kusto Query Language (KQL) to analyze usage and performance data?
9. What best practices should be followed when monitoring distributed applications?
10. How do you diagnose and resolve application performance bottlenecks using Azure telemetry?

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**1. What Azure tools are used to analyze application usage and performance metrics?**
*Azure Application Insights* and *Azure Monitor* are the primary tools. They provide telemetry on usage, performance, failures, and dependencies.

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**2. How do you enable application performance monitoring in Azure?**
Install and configure the *Application Insights SDK* or use the *Azure Monitor agent*. Enable monitoring from the *Azure portal* or through instrumentation in application code.

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**3. What types of application usage metrics can be collected in Azure?**
You can collect

- request counts,
- response times,
- failed requests,
- user sessions,
- page views,
- custom events,
- and dependency calls.

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**4. How do you visualize and interpret performance metrics in Application Insights?**
Use the *Application Insights* “Metrics” and “Performance” blades for charts and tables. Review response time distributions, failure trends, and drill into specific requests or dependencies.

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**5. How do you configure custom metrics and events in Application Insights?**
Track custom metrics or events using the *Application Insights SDK*.
Use methods like

- TrackEvent,
- TrackMetric,
- or TrackException

in your application code.

6. **What is the difference between availability, performance, and usage metrics?**

- Availability metrics show if an app is up and responding.
- Performance metrics measure speed (response times, resource usage).
- Usage metrics show how the app is used (sessions, users, page views).

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7. **How do you set up alerts based on application performance thresholds?**

In *Application Insights* or *Azure Monitor*, create an alert rule, define the target metric (e.g., response time > 1s), and assign an *Action Group* for notifications.

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8. **How can you use Kusto Query Language (KQL) to analyze usage and performance data?**

Use *Log Analytics* or *Application Insights* “Logs” to write KQL queries. Query tables like requests, dependencies, traces, and visualize results or export data.

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9. **What best practices should be followed when monitoring distributed applications?**

  - Instrument all services,
  - use distributed tracing,
  - correlate telemetry via operation IDs,
  - set up centralized dashboards
  - and alerts across services.

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11. **How do you diagnose and resolve application performance bottlenecks using Azure telemetry?**

  - Analyze slow requests and dependencies in *Application Insights*,
  - review end-to-end transaction views,
  - use failure and performance charts,
  - and drill into traces to find root causes.

Optimize code, resources, or dependencies based on findings.